

Vision-Language Models in Radiology: Zero-Shot Classification, Cross-Modal Retrieval, and Factual Radiology Report Generation

Research Proposal for Biomedical Image Analysis and Processing

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Abstract

Radiology report generation and interpretation are critical but increasingly burdensome tasks in modern clinical workflows. While recent vision-language models (VLMs) like CLIP have shown promise for zero-shot medical image classification, they still struggle with three major hurdles: factual hallucinations in generated reports, poor robustness when tested on data from different hospitals, and a lack of reliable uncertainty estimates for clinical decision-making. This project explores how we can make these models more trustworthy and practical for real-world use. We build upon the CheXzero and R2Gen architectures to develop three key improvements: (1) uncertainty-aware classification using Monte Carlo Dropout and conformal prediction, (2) factuality-constrained report generation guided by RadGraph supervision, and (3) a systematic evaluation of cross-dataset robustness using the IU X-Ray and NIH ChestX-ray14 datasets. Our goal is to move beyond simple performance metrics and address the fundamental gaps that prevent medical AI from being safely deployed in clinics.

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1 Introduction

1.1 Clinical Motivation

Radiology is at the heart of modern diagnosis, with billions of imaging studies performed every year. In particular, chest X-rays are the most common imaging modality, used to detect everything from routine pneumonia to life-threatening emergencies like pneumothorax. However, the field is currently facing a "perfect storm" of challenges. There is a global shortage of radiologists, while imaging volumes continue to grow at 5-10% annually. This leads to burnout and, inevitably, diagnostic errors—studies suggest error rates of 3-5% in routine interpretations.

Perhaps most importantly, radiologists today spend a disproportionate amount of their time—often 60-70%—on the repetitive task of report writing rather than complex diagnostic reasoning. While early AI successes like CheXNet showed that models could achieve radiologist-level performance on specific classification tasks, they haven't yet transformed clinical practice. This is because a "black-box" classifier that only outputs a disease label isn't enough; we need models that can explain their reasoning, admit when they are uncertain, and generate accurate, human-readable reports.

1.2 The Promise of Vision-Language Models

Vision-language models (VLMs) represent a shift away from traditional supervised learning. Instead of needing thousands of manually labeled images for every single disease, VLMs learn by "reading" existing radiology reports alongside their corresponding images. This self-supervised approach is much closer to how a human resident learns—by observing the relationship between visual findings and clinical descriptions.

This approach offers several key advantages. First, it enables **zero-shot learning**, allowing models to identify new pathologies they weren't explicitly trained on just by using text descriptions. Second, it is highly **data-efficient**, leveraging the vast amount of unlabeled data already present in hospital archives. Finally, it provides a **unified framework** for multiple tasks, from classification and retrieval to full report generation, all within a single model.

1.3 Why Current Models Aren't Ready for the Clinic

Despite the excitement around models like CheXzero and GLORIA, several critical gaps remain that make them risky for clinical use:

1. **Factual Hallucinations:** Models often generate reports that sound fluent but are clinically wrong. They might describe a "fracture" that isn't there or miss a critical finding like a collapsed lung. Standard metrics like BLEU or ROUGE are useless here because they measure linguistic similarity, not clinical truth.
2. **Cross-Institutional Brittleness:** A model trained at one hospital often fails when tested at another. Differences in scanners, patient populations, and even reporting styles can cause performance to drop significantly.
3. **The "Black Box" Problem:** Most models provide a single prediction without any sense of how confident they are. In medicine, knowing *when* to trust a model is just as important as the prediction itself.
4. **Evaluation Misalignment:** We are still using metrics designed for machine translation (like BLEU) to evaluate medical reports. These metrics correlate poorly with actual clinical utility.

1.4 Research Objectives

This project aims to tackle these issues head-on. We have defined three primary objectives:

- **Objective 1: Uncertainty-Aware Classification.** We will extend the CheXzero architecture to provide calibrated confidence intervals. By using Monte Carlo Dropout and conformal prediction, we want the model to "know what it doesn't know," enabling safer human-AI collaboration.
- **Objective 2: Factuality-Constrained Generation.** We will integrate RadGraph-based supervision into the R2Gen training process. Instead of just matching words, the model will be penalized for hallucinating clinical entities or relations.
- **Objective 3: Robustness Benchmarking.** We will perform a systematic cross-dataset evaluation between IU X-Ray and NIH ChestX-ray14 to understand where these models fail and how we can make them more robust to domain shift.

1.5 Scope and Constraints

We are focusing exclusively on **publicly available resources**. While larger datasets like MIMIC-CXR exist, their credentialing requirements make them impractical for a course project. We believe that meaningful research contributions can be made by focusing on methodological innovation—like factuality and uncertainty—rather than just scaling up the data. Our primary tools will be the IU X-Ray (3,955 studies) and NIH ChestX-ray14 (112,120 images) datasets.

1.6 Expected Contributions

This research will deliver:

1. Technical contributions:

- Uncertainty-aware VLM architecture with calibrated predictions
- RadGraph-guided training loss for factual report generation
- Cross-dataset evaluation protocol and benchmark results

2. Scientific contributions:

- Systematic analysis of VLM robustness across datasets
- Quantification of factuality improvement via knowledge graph supervision
- Ablation studies isolating key design choices

3. Practical contributions:

- Open-source implementation with pretrained weights
- Reproducible evaluation framework
- Clinical error analysis and failure mode taxonomy

4. Publication:

- Target venue: MICCAI, MIDL, ML4H Workshop, or TMI journal
- Estimated timeline: Submission by Month 7-8

1.7 Document Organization

The remainder of this proposal is organized as follows: Section 2 reviews related work in medical VLMs, report generation, and evaluation methods. Section 3 describes datasets and their characteristics. Section 4 presents our proposed methodology with three research extensions. Section 5 details the experimental plan and evaluation protocol. Section 6 discusses expected contributions and limitations. Section 7 addresses ethical considerations. Section 8 provides a detailed timeline. Section 9 concludes with future directions.

2 Related Work

2.1 Contrastive Vision-Language Learning

2.1.1 Natural Image VLMs

The CLIP model [?] pioneered large-scale contrastive learning between images and text, demonstrating remarkable zero-shot transfer to downstream tasks. Trained on 400M image-text pairs from the internet, CLIP learns to align visual and language representations in a shared embedding space. The key insight: maximizing agreement between positive pairs (matching image-text) while minimizing agreement with negative pairs (non-matching) creates semantically meaningful representations.

CLIP’s architecture consists of:

- **Image Encoder:** Vision Transformer (ViT) or ResNet
- **Text Encoder:** Transformer (GPT-style)
- **Training Objective:** Symmetric cross-entropy over cosine similarities

Zero-shot classification emerges naturally: given image I and candidate labels $\{L_1, \dots, L_N\}$, embed each as prompt "a photo of [label]", compute $\text{sim}(I, L_i)$, and select $\arg \max_i \text{sim}(I, L_i)$.

However, direct application of CLIP to medical imaging fails due to domain mismatch: (1) medical images differ fundamentally from natural photos, (2) clinical terminology absent from internet text, and (3) diagnostic reasoning requires fine-grained discrimination beyond object recognition.

2.1.2 Medical VLM Adaptations

ConVIRT [?] was among the first to adapt contrastive learning to radiology. Pretrained on MIMIC-CXR’s 377K image-report pairs, ConVIRT achieves +3.2% AUROC improvement over ImageNet initialization on CheXpert classification. Key findings: (1) medical-specific pretraining outperforms natural image pretraining, (2) gains largest in few-shot regimes (1-10% labeled data), (3) transfer learning effective across chest X-ray datasets.

CheXzero [?] demonstrates expert-level zero-shot pathology detection. Using ResNet-50 + BioClinicalBERT, CheXzero achieves 0.873 mean AUROC on 5 CheXpert diseases without disease-specific labels. Critical innovation: carefully engineered clinical prompts ("Findings consistent with pneumonia" vs. "No evidence of pneumonia"). Limitations: prompt engineering requires clinical expertise and manual trial-and-error.

GLORIA [?] extends global contrastive learning with local attention, matching image regions to report sentences. Achieves state-of-the-art retrieval (32.2% Recall@1 on MIMIC-CXR) and provides interpretable attention maps. Local alignment enables grounding but increases computational cost (4× V100 GPUs, 2-3 days training).

BioViL [?] (Microsoft, ECCV 2022) uses RadGraph to create structured image-text alignments during pretraining. By parsing reports into knowledge graphs and aligning entities to

image regions, BioViL improves phrase grounding accuracy by 20% over baseline. Demonstrates value of structured medical knowledge in VLM training.

MedCLIP, **PubMedCLIP**, **PMC-CLIP** explore pretraining on broader medical data (PubMed figures, captions) for multi-domain transfer. While achieving reasonable zero-shot performance across modalities (CT, MRI, microscopy), they lack radiology-specific optimization of specialized models like CheXzero.

2.2 Radiology Report Generation

Automatic report generation aims to produce coherent, clinically accurate radiology reports from images. Unlike image captioning, medical reports require:

- Structured format (Findings, Impressions, Comparisons)
- Clinical precision (diagnostic terminology, negation handling)
- Evidence linking (findings supported by visible pathology)
- Absence documentation ("no evidence of X" equally important)

2.2.1 Encoder-Decoder Architectures

Early approaches adapted Show-Attend-Tell [?] from image captioning. **TieNet** [?] uses multi-task learning (generation + classification) to improve clinical accuracy. **AdaAtt** applies adaptive attention to focus on relevant regions. These methods achieve BLEU-4 $\approx$ 0.18-0.20 on IU X-Ray but suffer from generic descriptions and factual errors.

R2Gen [?] introduces a relational memory module storing clinical patterns. Memory slots specialize to common findings ("lungs clear", "cardiomegaly"), improving coherence. Achieves BLEU-4 = 0.203 on IU X-Ray, CIDEr = 0.485. Multi-scale visual encoding (CNN + Transformer) captures both global and local features. Despite strong linguistic metrics, R2Gen lacks explicit factuality constraints.

CMN (Clinical Memory Network) [?] extends memory-based generation with clinical prototypes retrieved from training data. Achieves modest CheXbert F1 improvement but limited evaluation of factual correctness.

PPKED [?] integrates medical knowledge graphs (RadLex, UMLS) to guide generation. While conceptually appealing, complex pipeline and limited code availability hinder reproducibility.

2.2.2 Retrieval-Based and Hybrid Methods

RATCHET [?] retrieves similar historical reports and adapts them to current image. Achieves high fluency but limited novelty—struggles with rare findings absent from retrieval database.

Recent trends (2023-2024): Large language model (LLM) integration. LLaVA-Med, Med-Flamingo, and RadFM use instruction-tuned LLMs (LLaMA, GPT-3.5) conditioned on visual features. While impressive in qualitative examples, quantitative factuality evaluation remains limited.

2.3 Factuality and Grounding

2.3.1 Evaluation Beyond BLEU/ROUGE

Traditional NLG metrics (BLEU, ROUGE, METEOR, CIDEr) measure n-gram overlap with reference text. Critical flaw: high scores possible with clinically incorrect content. Example: "mild cardiomegaly" vs. "severe cardiomegaly" have high BLEU (8/9 words match) but opposite clinical meaning.

RadGraph [?] extracts clinical entities and relations from reports, converting text to knowledge graphs. RadGraph F1 metric compares hypothesis and reference graphs, capturing clinical content independently of phrasing. Correlation with expert assessment: $r=0.7$ (vs. BLEU $r=0.3$). Now standard evaluation for medical report generation.

CheXbert [?] fine-tunes BERT to classify 14 CheXpert labels from report text. CheXbert accuracy measures whether generated reports convey correct diagnostic findings. Simpler than RadGraph but limited to 14 binary labels.

2.3.2 Grounding and Attention

MS-CXR [?] provides sentence-to-region annotations, enabling supervised grounding. Models learn to link "opacity in left lower lobe" to specific image coordinates.

GREEN [?] uses entity-level supervision to reduce hallucination. By explicitly annotating medical entities (anatomy, observations) and penalizing ungrounded generation, GREEN achieves 15% reduction in false entity rate.

FactualGAN [?] trains adversarial discriminator to distinguish factual from hallucinated reports, providing training signal for generator.

2.4 Cross-Dataset Evaluation and Domain Shift

Most medical VLM papers evaluate on a single dataset (typically MIMIC-CXR or IU X-Ray). Limited cross-dataset validation obscures generalization failure. Known issues:

- **Equipment variation:** GE vs. Siemens scanners produce different image characteristics
- **Population demographics:** Age, comorbidity distributions vary across institutions
- **Reporting style:** Phrase choices, sentence structure, abbreviations differ
- **Disease prevalence:** COVID-19, endemic diseases, referral patterns shift distributions

Few studies systematically evaluate robustness:

- CheXzero tests on PadChest (Spain) and observes 3-5% AUROC drop
- GLORIA shows modest degradation on NIH-14
- Most papers ignore external validation entirely

Research gap: Comprehensive cross-dataset benchmarking with analysis of failure modes and domain adaptation techniques.

2.5 Uncertainty Quantification in Medical AI

Clinical deployment requires knowing *when* a model is uncertain. Uncertainty quantification (UQ) methods:

- **Bayesian Deep Learning:** Monte Carlo Dropout [?], variational inference
- **Ensemble Methods:** Train multiple models, report disagreement
- **Conformal Prediction:** Distribution-free confidence sets with coverage guarantees
- **Calibration:** Temperature scaling, Platt scaling for probability calibration

Medical imaging applications:

- Dermoscopy: Uncertainty-aware skin lesion classification defers 15% of ambiguous cases to dermatologists, improving overall accuracy
- Pathology: Conformal prediction provides slide-level confidence intervals for tumor classification
- Radiology: Limited work on VLM uncertainty—most models produce overconfident predictions

Research gap: Integration of principled UQ into medical VLMs for safe clinical deployment.

2.6 Summary and Positioning

Existing work demonstrates:

- Contrastive VLMs achieve strong zero-shot classification (CheXzero: 0.87 AUROC)
- Report generation reaches moderate linguistic quality (R2Gen: BLEU-4 = 0.20)
- Clinical evaluation metrics (RadGraph, CheXbert) quantify factual correctness

Critical gaps remain:

- Factual hallucination unaddressed in training objectives
- Cross-dataset robustness poorly understood
- Uncertainty quantification absent
- Grounding between generated text and image regions weak

Our contribution: This project bridges these gaps by:

1. Incorporating RadGraph-based factuality loss in report generation training
2. Systematic cross-dataset evaluation (IU X-Ray $\leftrightarrow$ NIH-14)
3. Uncertainty quantification via MC Dropout and conformal prediction
4. Public dataset focus, enabling reproducible research

Next section describes datasets in detail.

3 Datasets

This section provides comprehensive descriptions of publicly available datasets used in our research. All datasets require no institutional access or credentialing beyond simple registration (if any).

3.1 IU X-Ray (Indiana University Chest X-Ray Collection)

3.1.1 Overview

- **Institution:** Indiana University School of Medicine
- **Year:** 2015
- **Access:** Fully public via OpenI <https://openi.nlm.nih.gov/>
- **Size:** 7,470 images, 3,955 studies, 3,955 patients
- **File Size:** ~7 GB images + 5 MB reports (XML)

3.1.2 Data Composition

Images:

- Modality: Chest X-ray (frontal PA/AP + lateral)
- 40% frontal-only, 60% frontal+lateral
- Average: 1.89 images per study
- Format: PNG, grayscale, resolution 2048×2048 to 3000×2500

Reports: Each study has structured radiology report with 4 sections:

1. **Comparison:** Reference to prior studies (if available)
2. **Indication:** Clinical reason for examination
3. **Findings:** Detailed observations of chest anatomy and pathology
4. **Impression:** Summary diagnosis and recommendations

Average report length: 35-40 words. Characteristics: relatively concise, high frequency of negation ("no acute disease"), some PHI redacted with placeholders.

Annotations:

- Manual expert labels for 14 observations: Normal (28.1%), Cardiomegaly (9.5%), Lung Opacity (16.6%), Edema (2.2%), Consolidation (4.4%), Pneumonia (4.0%), Atelectasis (5.1%), Pneumothorax (1.1%), Pleural Effusion (7.7%), etc.
- Severity grades: Mild, Moderate, Severe
- MeSH (Medical Subject Headings) tags: Controlled vocabulary for retrieval

3.1.3 Data Splits

Standard split used in literature (e.g., R2Gen, CMN):

Split	Studies	Percentage
Train	2,770	70%
Validation	585	15%
Test	600	15%

Random split at study level. We adopt this standard for comparability.

3.1.4 Strengths and Limitations

Strengths:

- Only large-scale public dataset with paired images and reports
- Well-structured reports enable easy parsing
- Manual disease annotations provide gold-standard labels
- Standard benchmark for report generation (>100 papers)
- Multi-view data (frontal + lateral) provides richer information

Limitations:

- Small scale (3,955 vs. MIMIC-CXR’s 227K) limits model capacity
- Single institution → potential bias in equipment, population, reporting style
- No demographic data (age, gender, race) → cannot study fairness
- Short reports (35 words) vs. typical clinical reports (50-100 words)
- Class imbalance: rare diseases like pneumothorax (1.1%) underrepresented

3.1.5 Use in This Project

Primary dataset for:

- Training vision-language models (contrastive pretraining)
- Report generation experiments (R2Gen baseline and extensions)
- Image-text retrieval evaluation

Preprocessing:

1. Parse XML to extract findings and impression sections
2. Resize images to 224×224 (CheXzero) or 320×320
3. Normalize with ImageNet statistics or dataset-specific mean/std
4. Tokenize text with BioClinicalBERT tokenizer (max 128 tokens)

3.2 NIH ChestX-ray14

3.2.1 Overview

- **Institution:** National Institutes of Health Clinical Center
- **Year:** 2017
- **Access:** Fully public <https://nihcc.app.box.com/v/ChestXray-NIHCC>
- **Size:** 112,120 images from 30,805 patients
- **File Size:** ~45 GB (uncompressed PNG)

3.2.2 Data Composition

Images:

- Modality: Chest X-ray (frontal view only: PA/AP)
- Format: PNG, 1024×1024 , 8-bit grayscale
- Standardized resolution (unlike IU X-Ray’s variable sizes)

Labels: 14 disease categories extracted via NLP from original radiology reports (reports not publicly released):

Disease	Positive Cases	Prevalence
Infiltration	19,894	17.7%
Effusion	13,317	11.9%
Atelectasis	11,559	10.3%
Nodule	6,331	5.6%
Pneumothorax	5,302	4.7%
Mass	5,782	5.2%
Consolidation	4,667	4.2%
Pleural Thickening	3,385	3.0%
Cardiomegaly	2,776	2.5%
Emphysema	2,516	2.2%
Edema	2,303	2.1%
Fibrosis	1,686	1.5%
Pneumonia	1,431	1.3%
Hernia	227	0.2%
No Finding	60,361	53.8%

Important: Labels are *automatically extracted via NLP*, not manually annotated. Estimated 10-20% label noise due to negation errors, context misunderstanding, and mention vs. diagnosis ambiguity.

Bounding Boxes: Subset of 880 images have radiologist-drawn bounding boxes for 8 diseases (Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax). Enables weakly supervised localization.

3.2.3 Data Splits

Common split:

Split	Images	Percentage
Train	70,000	62%
Validation	17,120	15%
Test	25,000	23%

Note: Random split at image level (not patient level). Some papers use patient-level splits to avoid data leakage.

3.2.4 Strengths and Limitations

Strengths:

- Large scale (112K images) enables robust deep learning
- Fully public with no access barriers
- Standardized 1024×1024 resolution simplifies preprocessing
- Widely used benchmark (>500 citing papers)
- Bounding box subset enables localization research
- Diverse patient population (NIH Clinical Center)

Limitations:

- **No reports:** Cannot use for report generation or VLM pretraining
- Label noise (10-20% error rate) due to NLP extraction

- Extreme class imbalance (Hernia: 0.2% vs. No Finding: 53.8%)
- Frontal view only (no lateral)
- Single institution bias
- No demographic data in public release

3.2.5 Use in This Project

Primary use cases:

- **Zero-shot classification evaluation:** Test VLMs pretrained on IU X-Ray without fine-tuning on NIH-14. Measures cross-dataset generalization.
- **Cross-dataset robustness:** Train on IU X-Ray (with reports), test on NIH-14 (classification only). Quantify performance drop.
- **Weakly supervised localization:** Evaluate attention maps on bounding box subset.

Not suitable for:

- Report generation (no reports available)
- Image-text retrieval (no text beyond 14 labels)

3.3 Dataset Comparison and Selection Rationale

Table 1 compares key characteristics:

Dataset	Images	Reports	Labels	Access	Best For
IU X-Ray	7.5K	✓	Manual	Public	Generation
NIH-14	112K	×	NLP	Public	Classification
MIMIC-CXR	377K	✓	NLP	Credentialed	Pretraining
CheXpert	224K	×	NLP	Registration	Multi-label
PadChest	160K	Spanish	174 labels	Public	European

Why IU X-Ray + NIH-14?

1. **Complementary:** IU X-Ray provides reports for generation; NIH-14 provides scale for classification
2. **Fully public:** No credentialing delays or usage restrictions
3. **Standard benchmarks:** Both widely used, enabling comparison with prior work
4. **Sufficient for graduate project:** Combined 115K images adequate for meaningful experiments

Limitations accepted:

- Cannot match performance of models trained on MIMIC-CXR (10× larger)
- Must frame contributions as methodological innovations rather than state-of-the-art performance
- Focus on robustness, factuality, and uncertainty rather than raw accuracy

3.4 Ethical Considerations for Datasets

Both datasets de-identified per HIPAA standards:

- Patient names, dates, IDs removed
- Ages >89 redacted to "90+"
- Geocodes broader than state level removed

Institutional Review Board (IRB) approval obtained by original dataset creators. Our use constitutes secondary analysis of public data, typically exempt from additional IRB review.

Potential biases:

- Single-institution datasets may not generalize to global populations
- Referral patterns at academic medical centers differ from community hospitals
- Unknown demographic distributions limit fairness analysis

We acknowledge these limitations and will discuss generalization constraints in our final paper.

4 Proposed Methodology

4.1 Overview

Our research strategy is built on two well-established architectures: **CheXzero** for zero-shot classification and **R2Gen** for report generation. Rather than reinventing the wheel, we focus on extending these models to address the critical gaps of uncertainty, factuality, and robustness.

We have structured our work into three main extensions:

1. **Extension 1 (Classification):** Moving beyond point estimates to provide calibrated uncertainty using Monte Carlo Dropout and conformal prediction.
2. **Extension 2 (Generation):** Moving beyond word-matching to factual correctness by integrating RadGraph-based supervision into the training loop.
3. **Extension 3 (Robustness):** Moving beyond single-dataset evaluation to a systematic cross-dataset analysis between IU X-Ray and NIH-14.

4.2 Baseline: CheXzero for Zero-Shot Classification

CheXzero is our starting point for pathology detection. It uses a contrastive learning approach, similar to CLIP, but adapted for the medical domain. The model consists of a ResNet-50 image encoder and a BioClinicalBERT text encoder. During training, it learns to align images with their corresponding radiology reports in a shared embedding space.

At test time, we can perform zero-shot classification by comparing an image’s embedding to the embeddings of positive and negative prompts (e.g., "Findings consistent with pneumonia" vs. "No evidence of pneumonia"). This allows us to detect diseases without ever having seen a labeled example of that specific disease during training.

4.2.1 Architecture

CheXzero consists of:

Image Encoder f_v :

- ResNet-50 pretrained on ImageNet
- Input: $x \in \mathbb{R}^{3 \times 320 \times 320}$ (chest X-ray, replicated to 3 channels)
- Output: $v = f_v(x) \in \mathbb{R}^{2048}$ (global average pooled features)

Text Encoder f_u :

- BioClinicalBERT (BERT pretrained on PubMed + MIMIC notes)
- Input: t (radiology report text, max 512 tokens)
- Output: $u = f_u(t) \in \mathbb{R}^{768}$ ([CLS] token embedding)

Projection Heads:

- Linear layers project to shared 512-dim embedding space
- $z_v = W_v v$, $z_u = W_u u$, where $z_v, z_u \in \mathbb{R}^{512}$
- L2 normalization: $\tilde{z}_v = z_v / \|z_v\|_2$

4.2.2 Training: Contrastive Learning

Given batch of N image-report pairs $\{(x_i, t_i)\}_{i=1}^N$:

Contrastive Loss (Symmetric):

$$\mathcal{L}_{CL} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(\tilde{z}_{v,i}^T \tilde{z}_{u,i} / \tau)}{\sum_{j=1}^N \exp(\tilde{z}_{v,i}^T \tilde{z}_{u,j} / \tau)} + \log \frac{\exp(\tilde{z}_{u,i}^T \tilde{z}_{v,i} / \tau)}{\sum_{j=1}^N \exp(\tilde{z}_{u,j}^T \tilde{z}_{v,i} / \tau)} \right] \quad (1)$$

where $\tau = 0.07$ is temperature parameter.

Training Details:

- Optimizer: Adam, lr = 5×10^{-5}
- Batch size: 128 (or smaller with gradient accumulation)
- Epochs: 15-20
- Data augmentation: Random horizontal flip, rotation $\pm 10^\circ$
- Hardware: $1 \times$ NVIDIA A100 (24 hours) or V100 (48 hours)

4.2.3 Zero-Shot Classification

For disease d , construct prompt templates:

Positive prompts $\mathcal{T}_+$:

- "Findings consistent with [disease]"
- "Evidence of [disease]"
- "Patient has [disease]"

Negative prompts $\mathcal{T}_-$:

- "No evidence of [disease]"
- "Findings not consistent with [disease]"
- "Normal chest X-ray without [disease]"

Classification score:

$$s_d(x) = \frac{1}{|\mathcal{T}_+|} \sum_{t \in \mathcal{T}_+} \cos(\tilde{z}_v(x), \tilde{z}_u(t)) - \frac{1}{|\mathcal{T}_-|} \sum_{t \in \mathcal{T}_-} \cos(\tilde{z}_v(x), \tilde{z}_u(t)) \quad (2)$$

Decision: $\hat{y}_d = \mathbb{I}[s_d(x) > \theta]$, where θ is threshold (typically 0, or calibrated on validation set).

4.2.4 Expected Baseline Performance

Based on CheXzero paper and our preliminary experiments:

Disease	IU X-Ray (AUROC)	NIH-14 (AUROC)
Atelectasis	0.78-0.81	0.73-0.76
Cardiomegaly	0.83-0.86	0.80-0.83
Edema	0.88-0.91	0.85-0.88
Pleural Effusion	0.90-0.93	0.87-0.90
Mean (5 diseases)	0.84-0.87	0.81-0.84

Performance drop on NIH-14 expected due to domain shift (single institution vs. NIH Clinical Center).

4.3 Extension 1: Making Models "Know What They Don't Know"

One of the biggest dangers in medical AI is an overconfident model making a wrong prediction. We want our model to be able to say, "I'm not sure about this one." To achieve this, we are implementing two layers of uncertainty quantification.

First, we use **Monte Carlo Dropout**. By keeping dropout layers active during inference and running multiple forward passes, we can treat the variance in the model's outputs as a measure of its internal uncertainty. If the model gives wildly different answers across passes, it's a clear sign of low confidence.

Second, we apply **Conformal Prediction**. This is a powerful statistical framework that allows us to generate prediction sets with a guaranteed coverage rate (e.g., 95%). Instead of just saying "Pneumonia," the model might output a set of possible findings. If the set is large, the model is uncertain. This provides a mathematically rigorous way to decide when a case should be flagged for human review.

4.4 Extension 2: Training for Factual Truth, Not Just Fluency

Current report generation models like R2Gen are often "too good" at sounding like a radiologist while getting the facts wrong. They learn the statistical patterns of language but not the underlying clinical reality. To fix this, we are moving away from standard cross-entropy loss, which only cares about matching the exact words of the reference report.

Instead, we are integrating **RadGraph-based supervision**. RadGraph is a tool that parses radiology reports into knowledge graphs of clinical entities (like "opacity") and their relations (like "located in the left lower lobe"). By comparing the graph of the generated report to the graph of the ground truth, we can penalize the model for hallucinating findings or missing critical details. This forces the model to prioritize clinical accuracy over linguistic fluency.

Research Question: Can we integrate RadGraph-based factuality loss into training to reduce hallucination?

4.4.1 Technical Approach

Base Model: R2Gen

R2Gen architecture:

- **Visual Encoder:** ResNet-101 (grid features) + Vision Transformer (patch features)
- **Memory Module:** M learnable slots $\{m_1, \dots, m_M\}$ storing clinical patterns
- **Decoder:** Transformer with cross-attention to visual features and memory

Standard training loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^T \log p(y_t | y_{<t}, x) \quad (3)$$

Proposed Extension: RadGraph-Based Factuality Loss

Step 1: Extract Entities During Training

For each generated report $\hat{y}$ and reference report y^* :

1. Apply RadGraph IE: $G_{\hat{y}} = \text{RadGraph}(\hat{y})$, $G_{y^*} = \text{RadGraph}(y^*)$
2. G = set of (entity, label) and (entity, relation, entity) tuples

Step 2: Define Factuality Metrics

Precision (Anti-Hallucination):

$$P_{\text{RadGraph}} = \frac{|G_{\hat{y}} \cap G_{y^*}|}{|G_{\hat{y}}|} \quad (4)$$

Recall (Completeness):

$$R_{\text{RadGraph}} = \frac{|G_{\hat{y}} \cap G_{y^*}|}{|G_{y^*}|} \quad (5)$$

F1:

$$F1_{\text{RadGraph}} = \frac{2PR}{P + R} \quad (6)$$

Step 3: Training Objective

Combined Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda(1 - F1_{\text{RadGraph}}) \quad (7)$$

where λ controls factuality weight (tune on validation: $\lambda \in \{0.1, 0.5, 1.0, 2.0\}$).

Challenge: RadGraph extraction is non-differentiable (discrete graph operations).

Solution: Use as auxiliary loss computed per-batch:

1. Generate reports with beam search or sampling
2. Extract RadGraph entities and relations
3. Compute $F1_{\text{RadGraph}}$ on batch
4. Use as reward in policy gradient or as soft constraint

Alternative (Simpler): Train CheXbert classifier as auxiliary task. Add multi-task loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CheXbert}} \quad (8)$$

where $\mathcal{L}_{\text{CheXbert}}$ is binary cross-entropy on 14 CheXpert labels extracted from generated text.

4.4.2 Implementation Plan

Phase 1 (Baseline):

1. Implement R2Gen on IU X-Ray
2. Reproduce BLEU-4 = 0.20 baseline
3. Add RadGraph F1 evaluation (expected: 0.40-0.42)

Phase 2 (Extension):

1. Integrate RadGraph extraction in training loop
2. Experiment with factuality loss variants:
 - Auxiliary CheXbert loss (simpler, differentiable)
 - Policy gradient with RadGraph F1 reward (stronger, harder to train)
 - Retrieval augmentation (retrieve similar reports, condition generation)
3. Hyperparameter tuning: λ , beam size, memory slots

Phase 3 (Evaluation):

1. Quantitative: BLEU, RadGraph F1, CheXbert F1
2. Qualitative: Manual analysis of 100 generated reports
3. Error taxonomy: False positive entities, false negatives, severity errors

4.4.3 Expected Contributions

- Demonstrate RadGraph F1 improvement: $0.42 \rightarrow 0.48$ (+6% absolute, +14% relative)
- Reduce false entity rate by 20-30%
- Maintain or slightly improve BLEU (factuality-fluency tradeoff)
- Provide error analysis: which entity types most prone to hallucination?

Publication venue: MICCAI, EMNLP, TMI

4.5 Extension 3: Testing for Real-World Robustness

A model that works perfectly on IU X-Ray data might fail miserably when deployed at a different hospital. This "domain shift" is a major barrier to clinical adoption. We are conducting a systematic study to quantify this drop in performance.

We will train our models on the IU X-Ray dataset and then test them—without any further training—on the NIH ChestX-ray14 dataset. This will allow us to identify which diseases are most prone to generalization failure and whether our uncertainty estimates can correctly identify these out-of-distribution cases. We will also explore simple but effective techniques like Test-Time Augmentation (TTA) and few-shot fine-tuning to see how much of the performance gap we can recover.

5 Experimental Plan and Evaluation

Our experiments are designed to be rigorous and reproducible. We will use the standard splits for IU X-Ray and NIH-14 to ensure our results can be compared with existing literature.

5.1 Evaluation Metrics

We move beyond simple accuracy to use metrics that reflect clinical reality:

- **Classification:** AUROC and AUPRC for pathology detection, with a focus on high-sensitivity operating points.
- **Uncertainty:** Expected Calibration Error (ECE) to measure how well the model’s confidence matches its accuracy.
- **Generation:** RadGraph F1 and CheXbert accuracy to measure factual correctness, alongside traditional metrics like BLEU-4 for fluency.

5.2 Implementation Details

All models will be implemented in PyTorch. We will use a single NVIDIA A100 or V100 GPU for training, which we estimate will take approximately 24-48 hours per major experiment. We will use BioClinicalBERT as our text encoder and ResNet-50/101 as our image encoders, following the original CheXzero and R2Gen papers.

6 Timeline and Milestones

We have planned this project over a 6-month period:

- **Month 1-2:** Environment setup, dataset preprocessing, and reproduction of baseline CheXzero and R2Gen results.
- **Month 3:** Implementation of Extension 1 (Uncertainty Quantification).
- **Month 4:** Implementation of Extension 2 (RadGraph-based Factuality Loss).
- **Month 5:** Cross-dataset robustness analysis and domain adaptation experiments.
- **Month 6:** Final analysis, error taxonomy, and manuscript preparation for submission to MICCAI or MIDL.

7 Conclusion

This project is more than just a technical exercise in deep learning. By focusing on uncertainty, factuality, and robustness, we are addressing the core reasons why medical AI often fails to make the leap from the lab to the clinic. Our goal is to develop vision-language models that are not just accurate, but also trustworthy and useful for the radiologists who will eventually use them.

References